

Day 8 Student Handout

Geometry Summer Credit | Similarity, Scale Factor, and AA

Name:	Date:	Period:
-------	-------	---------

Learning Targets

I can identify similar figures using scale factor and dilations. I can use AA similarity to decide when triangles are similar and solve proportions for missing side lengths.

Lesson 8A: Similarity Transformations and Scale Factor

Warm-Up

1.	Solve the proportion: $\frac{3}{5} = \frac{x}{20}$.
2.	A rectangle is 4 by 7. A scaled copy has corresponding sides 12 and 21. What is the scale factor?
3.	A dilation centered at the origin with scale factor 3 maps A(2, -5) to what point?

Concept Launch

Similar Means Same Shape

Similar figures have the same shape, but not necessarily the same size. A dilation changes size by a scale factor. Rigid motions may move or flip the figure, but they do not change size.

Problem	Work / notes
Triangle ABC has side lengths 5, 8, 10. Triangle DEF has side lengths 15, 24, 30. Are they similar? What is the scale factor from ABC to DEF?	Ratios: Scale factor: Conclusion:
A figure is dilated by scale factor $\frac{1}{2}$. What happens to each side length?	Original side: New side: Relationship:
Point P(-4, 6) is dilated by scale factor 2 about the origin. Find P'.	Rule: Image point:

A scale drawing uses 1 inch = 6 feet. A wall is 4 inches on the drawing. What is the actual length?

Scale factor:
Actual length:

Practice

4. A side length of 9 is scaled by factor 4. What is the new side length?

5. A model car is 7 inches long. The real car is 14 feet long. If 14 feet = 168 inches, what is the scale factor from model to real?

6. Dilation rule: $(x, y) \rightarrow (0.5x, 0.5y)$. Find the image of $Q(10, -6)$.

Day 8 Student Handout

Lesson 8B | AA Similarity and Proportions

Lesson 8B: AA Similarity

AA Similarity

If two angles in one triangle are congruent to two angles in another triangle, the triangles are similar. Once triangles are similar, corresponding side lengths are proportional.

Problem	Work / notes
Triangle ABC has angles 40, 65, 75. Triangle DEF has angles 40, 75, 65. Are the triangles similar?	Angle matches: Conclusion:
Triangle ABC is similar to triangle XYZ. $AB = 6$, $XY = 18$, $BC = 8$. Find YZ .	Scale factor: Proportion: Answer:
Triangle JKL is similar to triangle MNP. $JK = 10$, $MN = 25$, $KL = x$, $NP = 30$. Find x .	Corresponding sides: Proportion: Answer:
A student sets up $4/10 = 12/x$ from similar triangles. Solve for x .	Cross multiply: Answer:

Practice

7.	Are two triangles with angle pairs 52 degrees and 68 degrees similar? Explain.
8.	Triangle ABC is similar to triangle DEF. $AB = 5$, $DE = 15$, $BC = 9$. Find EF .
9.	Solve: $7/12 = x/36$.
10.	Explain the difference between congruent and similar figures.